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Branch – AI&DS

# Experiment-05

AIM: To understand Continuous integration, Jenkins installation.

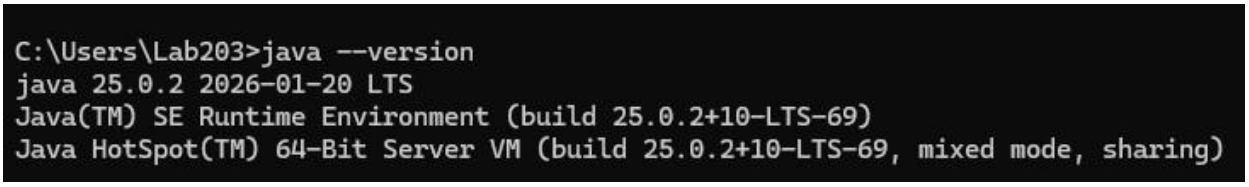
LO mapped: LO3

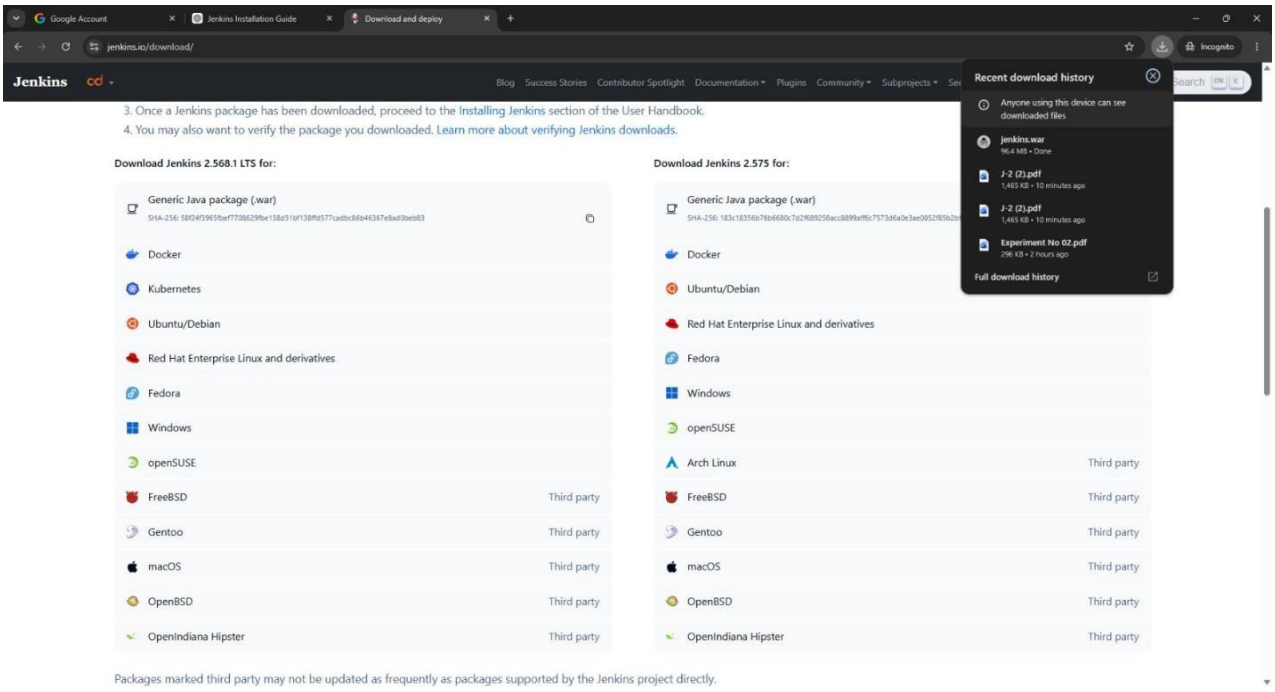
Theory:

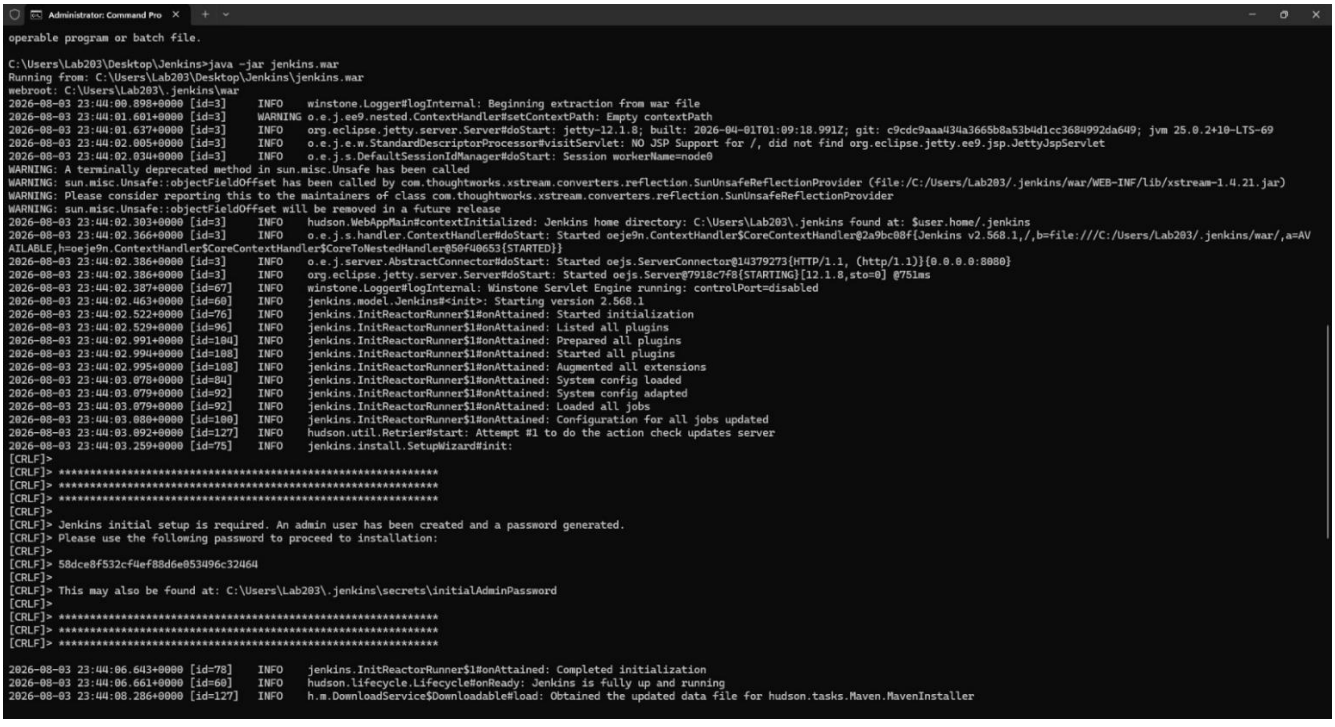
Continuous Integration is a development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

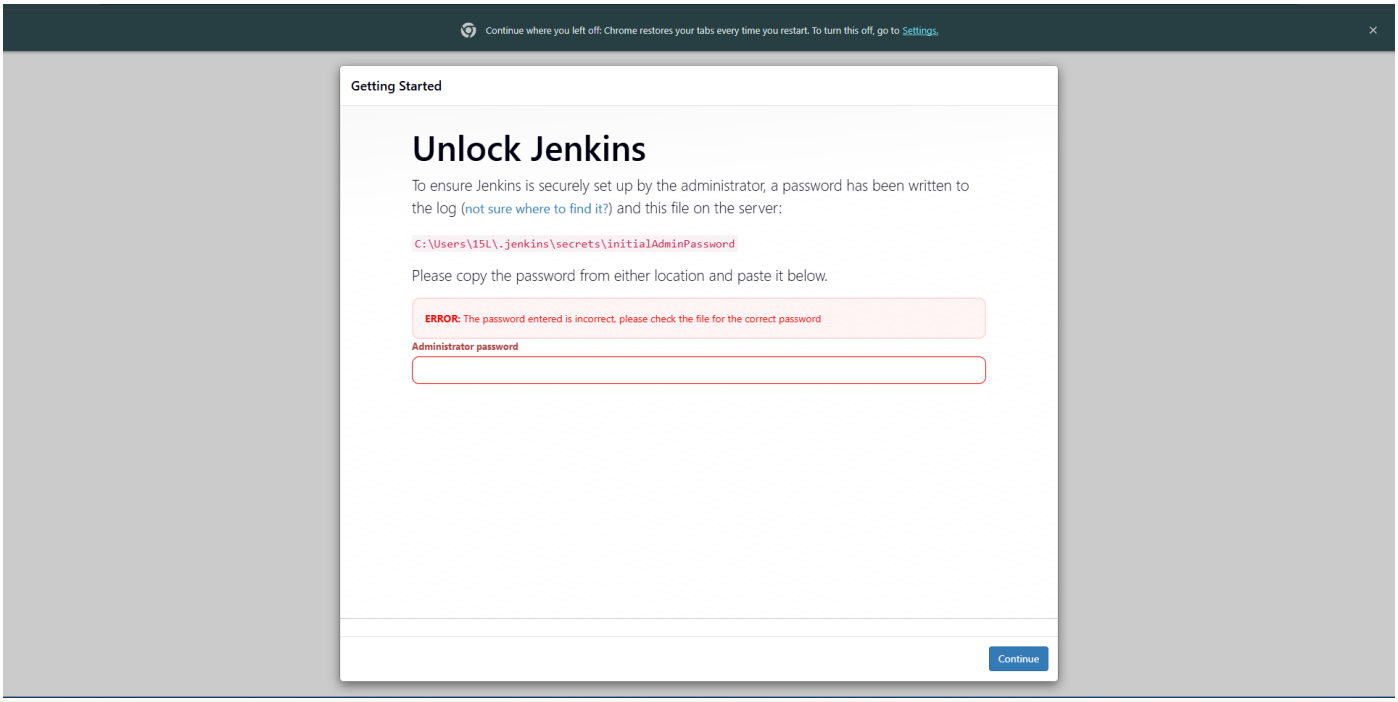
Jenkins is an open-source automation server that facilitates continuous integration and continuous delivery (CI/CD) of software projects. It helps developers automate various stages of the software development lifecycle, from building and testing to deployment and monitoring. Jenkins supportsa wide range of plugins and integrations, making it a versatile tool for automating various tasks in software development.

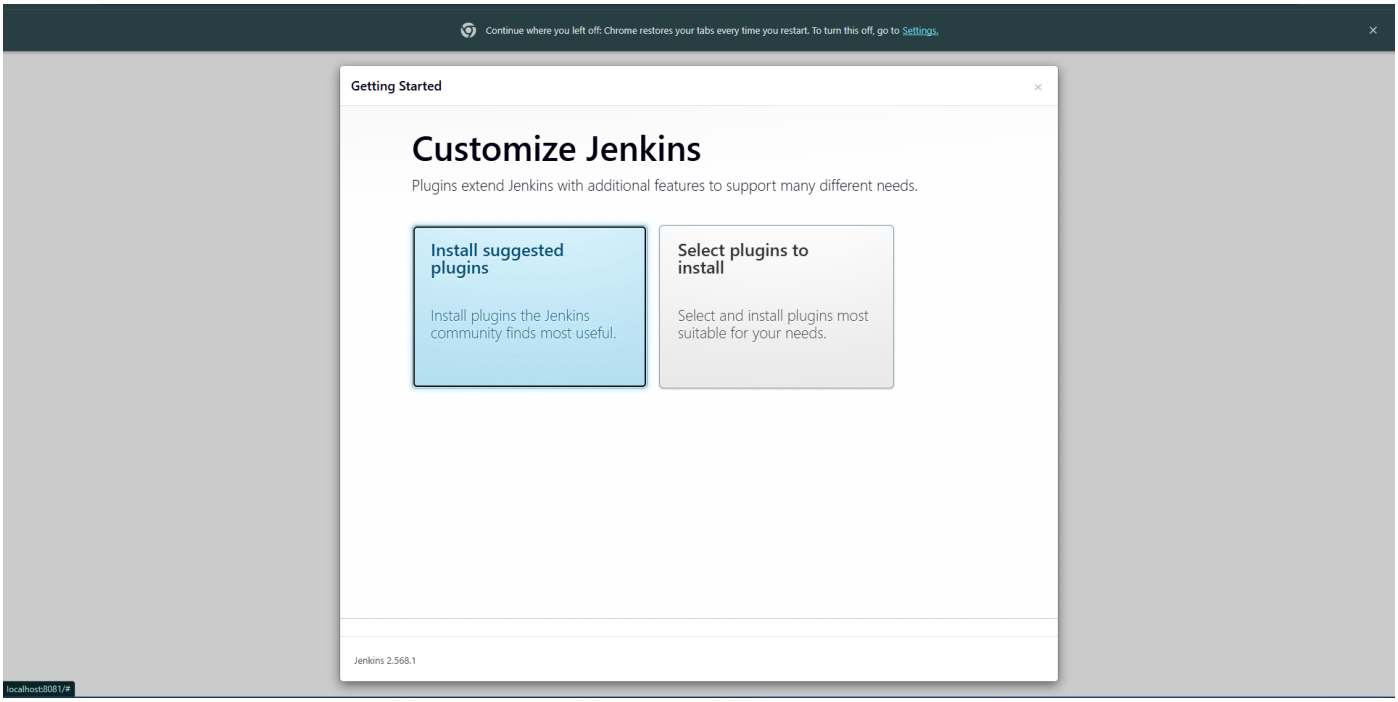
OUTPUT:

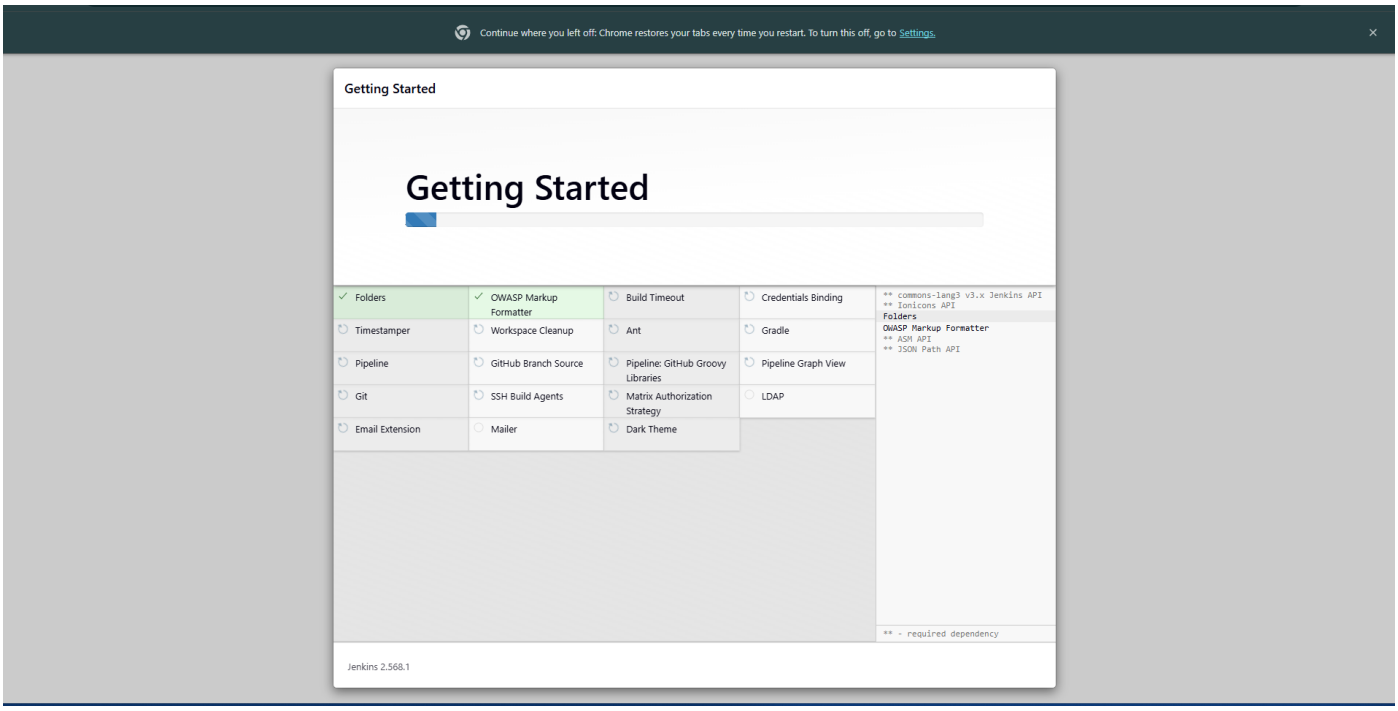


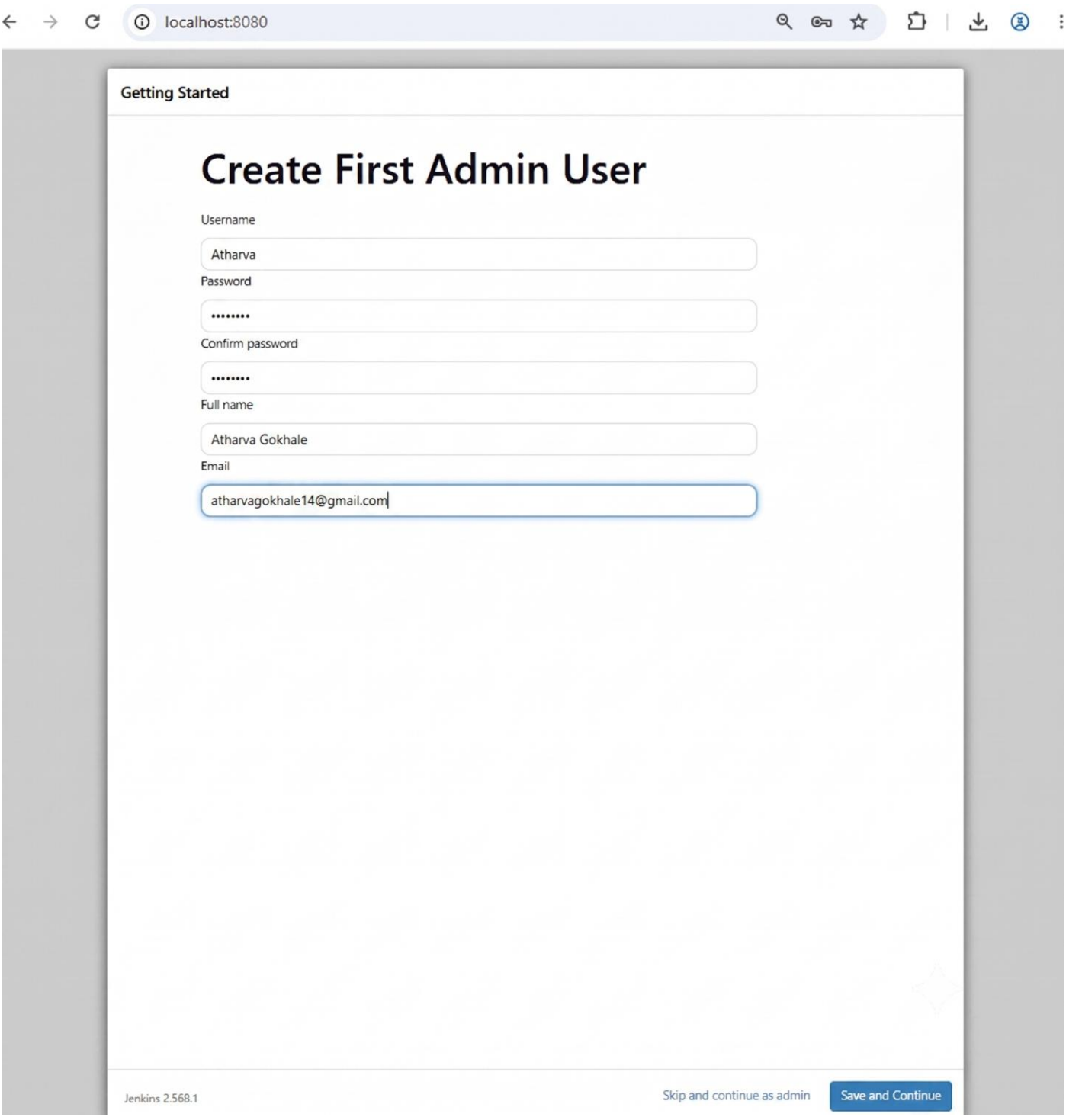

Google Account
Jenkins Installation Guide
Download and deploy
jenkins.io/download/
Incognito
Jenkins
Blog
Success Stories
Contributor Spotlight
Documentation
Plugins
Community
Subprojects
Search
Recent download history
Anyone using this device can see downloaded files
jenkins.war
96.4 MB • Done
J-2 (2).pdf
1,465 KB • 10 minutes ago
J-2 (2).pdf
1,465 KB • 10 minutes ago
Experiment No 02.pdf
296 KB • 2 hours ago
Full download history
3. Once a Jenkins package has been downloaded, proceed to the Installing Jenkins section of the User Handbook.
4. You may also want to verify the package you downloaded. Learn more about verifying Jenkins downloads.
Download Jenkins 2.568.1 LTS for:
Generic Java package (.war)
Docker
Kubernetes
Ubuntu/Debian
Red Hat Enterprise Linux and derivatives
Fedora
Windows
openSUSE
FreeBSD
Third party
Gentoo
Third party
macOS
Third party
OpenBSD
Third party
OpenIndiana Hipster
Third party
Download Jenkins 2.575 for:
Generic Java package (.war)
Docker
Ubuntu/Debian
Red Hat Enterprise Linux and derivatives
Fedora
Windows
openSUSE
Arch Linux
Third party
FreeBSD
Third party
Gentoo
Third party
macOS
Third party
OpenBSD
Third party
OpenIndiana Hipster
Third party
Packages marked third party may not be updated as frequently as packages supported by the Jenkins project directly.

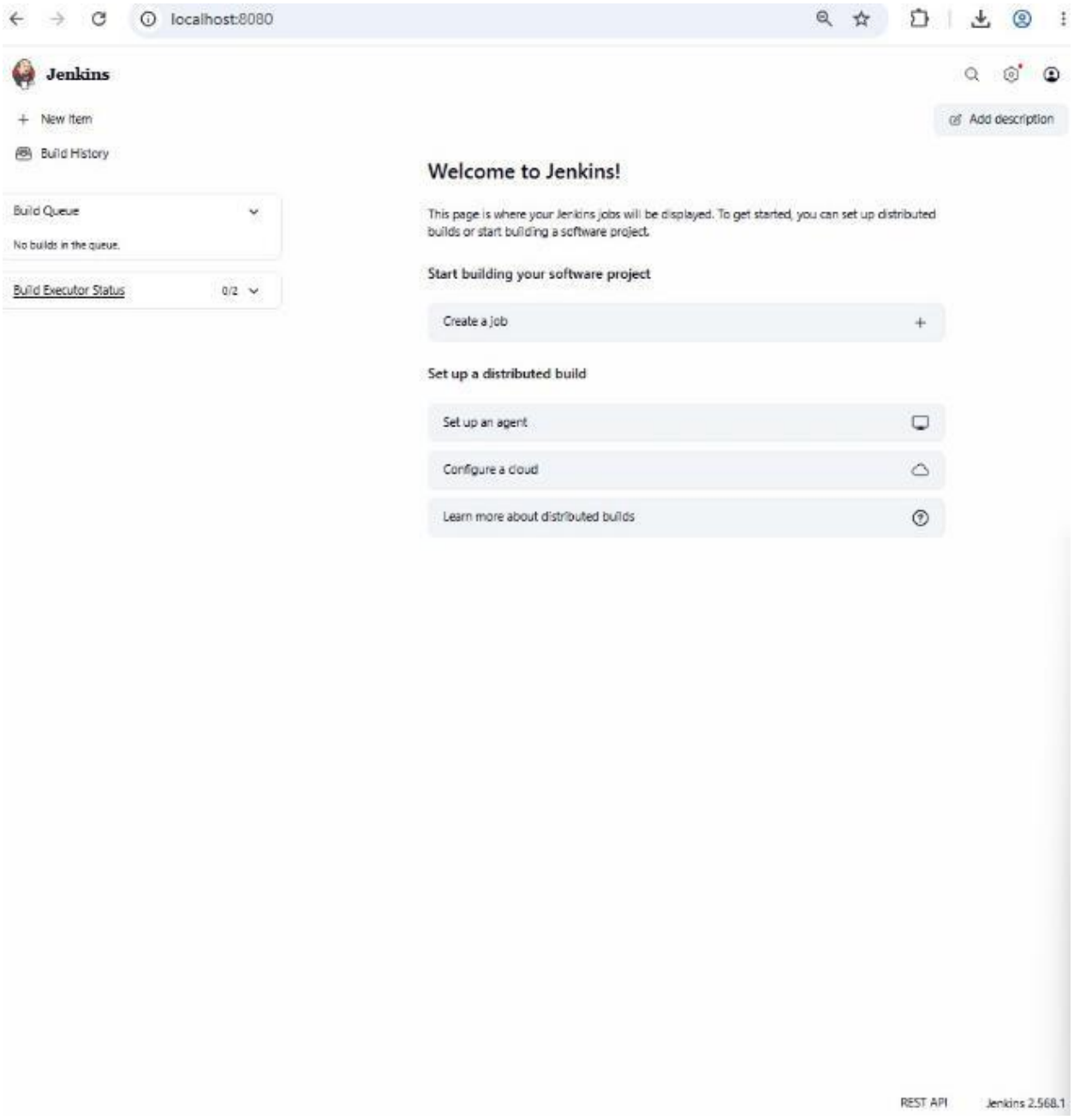

Administrator: Command Pro
operable program or batch file.
C:\Users\Lab203\Desktop\Jenkins>java -jar jenkins.war
Running from: C:\Users\Lab203\Desktop\Jenkins\jenkins.war
webroot: C:\Users\Lab203\.jenkins\war
2026-08-03 23:44:00.898+0000 [id=3] INFO winstone.Logger#logInternal: Beginning extraction from war file
2026-08-03 23:44:01.601+0000 [id=3] WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-08-03 23:44:01.637+0000 [id=3] INFO org.eclipse.jetty.server.Server#doStart: jetty-12.1.8; built: 2026-04-01T01:09:18.991Z; git: c9cdc9aaa434a3665b8a53b4d1cc3684992da649; jvm 25.0.2+10-LTS-69
2026-08-03 23:44:02.005+0000 [id=3] INFO o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-08-03 23:44:02.034+0000 [id=3] INFO o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider (file:/C:/Users/Lab203/.jenkins/war/WEB-INF/lib/xstream-1.4.21.jar)
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
2026-08-03 23:44:02.303+0000 [id=3] INFO hudson.WebAppMain#contextInitialized: Jenkins home directory: C:\Users\Lab203\.jenkins found at: $user.home/.jenkins
2026-08-03 23:44:02.366+0000 [id=3] INFO o.e.j.s.handler.ContextHandler#doStart: Started oeje9n.ContextHandler$CoreContextHandler@2a9bc08f{Jenkins v2.568.1,/,b=file:///C:/Users/Lab203/.jenkins/war/,a=AVAILABLE,h=oeje9n.ContextHandler$CoreContextHandler$CoreToNestedHandler@50f40653{STARTED}}
2026-08-03 23:44:02.386+0000 [id=3] INFO o.e.j.server.AbstractConnector#doStart: Started oejs.ServerConnector@14379273{HTTP/1.1, (http/1.1)}{0.0.0.0:8080}
2026-08-03 23:44:02.386+0000 [id=3] INFO org.eclipse.jetty.server.Server#doStart: Started oejs.Server@7918c7f8{STARTING}[12.1.8,sto=0] @751ms
2026-08-03 23:44:02.387+0000 [id=67] INFO winstone.Logger#logInternal: Winstone Servlet Engine running: controlPort=disabled
2026-08-03 23:44:02.463+0000 [id=60] INFO jenkins.model.Jenkins#<init>: Starting version 2.568.1
2026-08-03 23:44:02.522+0000 [id=76] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-03 23:44:02.529+0000 [id=96] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-03 23:44:02.991+0000 [id=104] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-03 23:44:02.994+0000 [id=108] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-03 23:44:02.995+0000 [id=108] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-03 23:44:03.070+0000 [id=84] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-03 23:44:03.079+0000 [id=92] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-03 23:44:03.079+0000 [id=92] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-03 23:44:03.088+0000 [id=100] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-03 23:44:03.092+0000 [id=127] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
2026-08-03 23:44:03.259+0000 [id=75] INFO jenkins.install.SetupWizard#init:
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> 58dce8f532cf4ef88d6e053496c32464
[CRLF]>
[CRLF]> This may also be found at: C:\Users\Lab203\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> ************************************************************
[CRLF]> ************************************************************
[CRLF]> ************************************************************
2026-08-03 23:44:06.643+0000 [id=78] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-03 23:44:06.661+0000 [id=60] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-08-03 23:44:08.286+0000 [id=127] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller

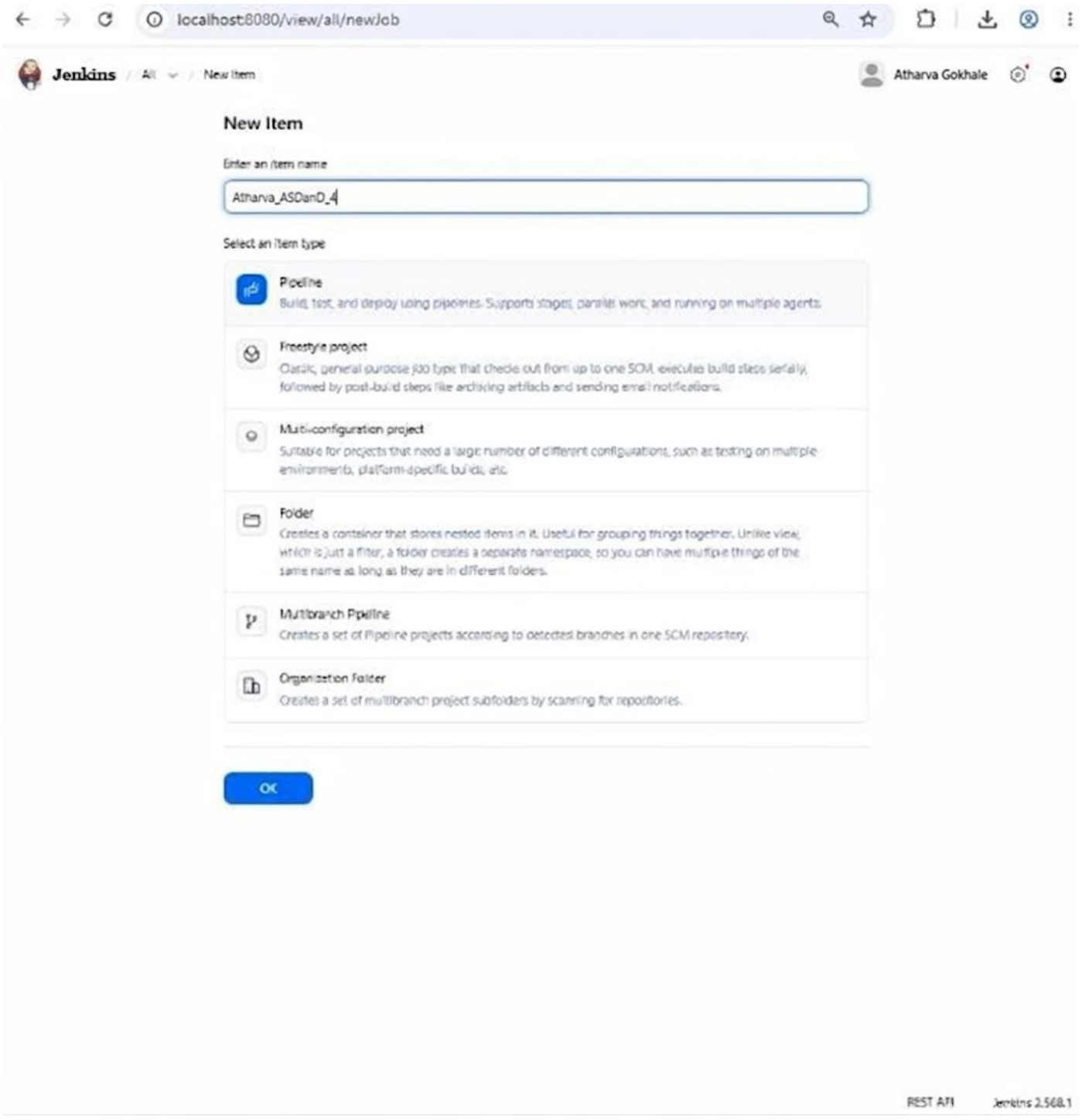

Continue where you left off: Chrome restores your tabs every time you restart. To turn this off, go to Settings.
Getting Started
Unlock Jenkins
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:
C:\Users\15L\.jenkins\secrets\initialAdminPassword
Please copy the password from either location and paste it below.
ERROR: The password entered is incorrect, please check the file for the correct password
Administrator password
Continue

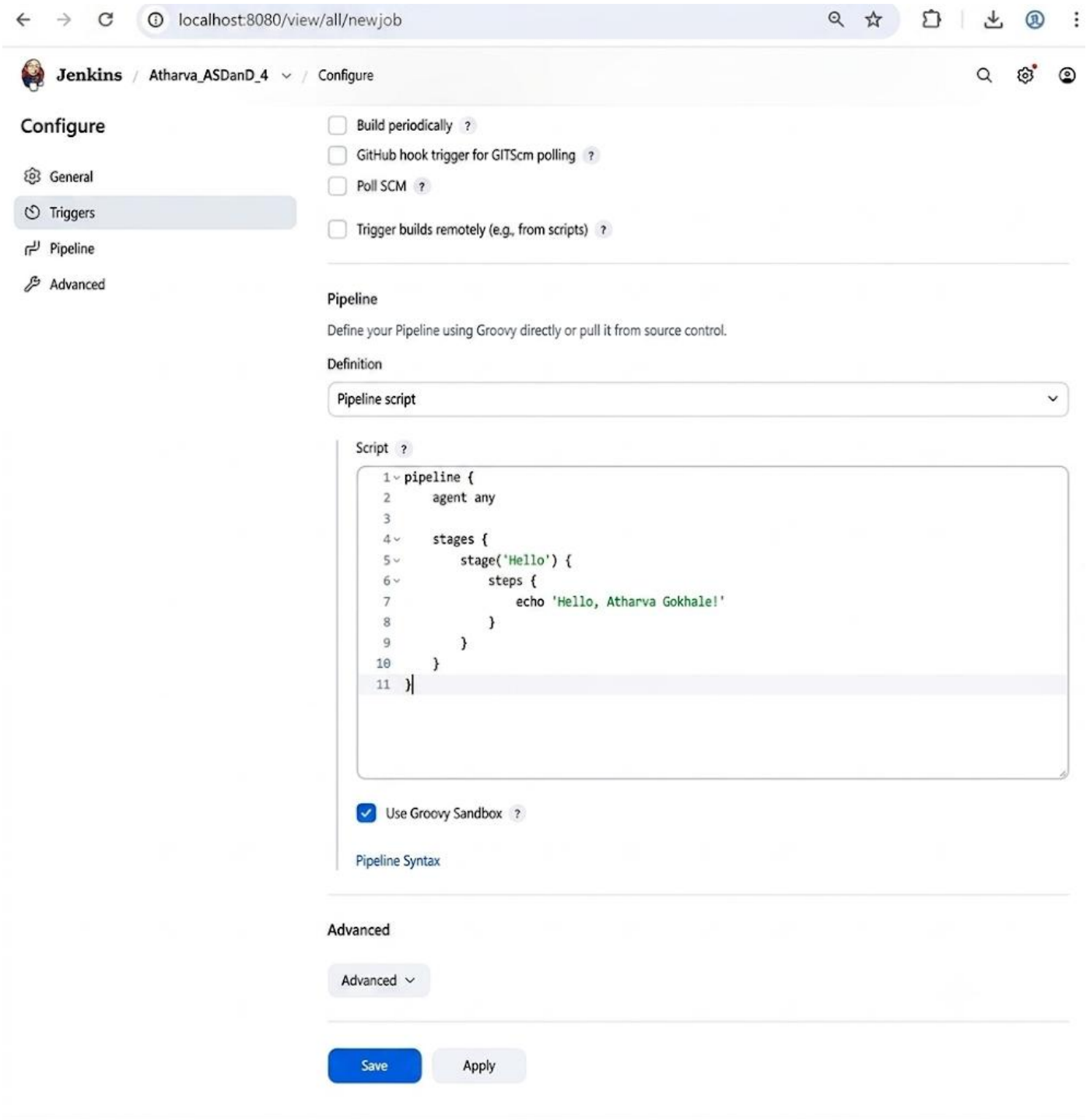

Continue where you left off: Chrome restores your tabs every time you restart. To turn this off, go to Settings.
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.568.1
localhost:8081/#

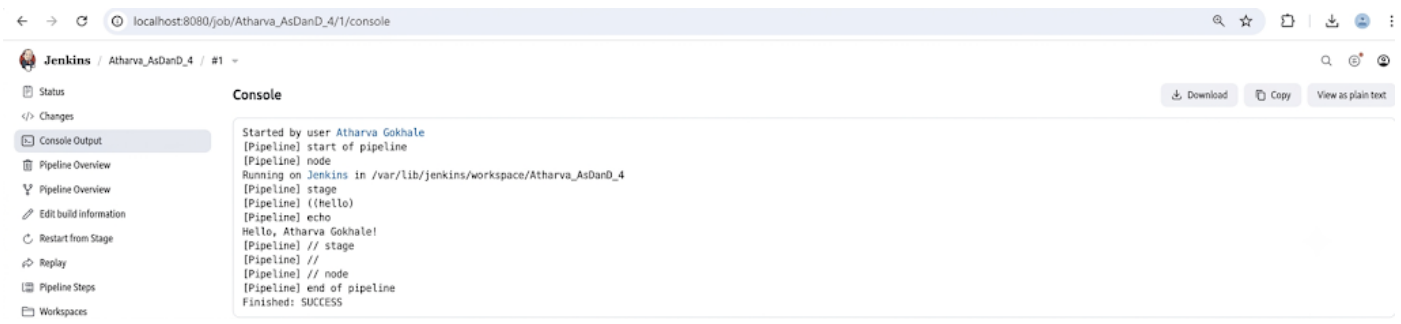

Continue where you left off: Chrome restores your tabs every time you restart. To turn this off, go to Settings.
Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** JSON Path API
** - required dependency
Jenkins 2.568.1


localhost:8080
Getting Started
Create First Admin User
Username
Atharva
Password
Confirm password
Full name
Atharva Gokhale
Email
atharvagokhale14@gmail.com
Jenkins 2.568.1
Skip and continue as admin
Save and Continue


localhost:8080
Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status
0/2
Add description
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
Learn more about distributed builds
REST API
Jenkins 2.568.1


localhost:8080/view/all/newJob
Jenkins
All
New Item
Atharva Gokhale
New Item
Enter an item name
Atharva_ASDanD_4
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
OK
REST API
Jenkins 2.568.1


localhost:8080/view/all/newjob
Jenkins / Atharva_ASDanD_4 / Configure
Configure
General
Triggers
Pipeline
Advanced
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Trigger builds remotely (e.g., from scripts)
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
pipeline {
agent any
stages {
stage('Hello') {
steps {
echo 'Hello, Atharva Gokhale!'
}
}
}
}
Use Groovy Sandbox
Pipeline Syntax
Advanced
Advanced
Save
Apply


localhost:8080/job/Atharva_AsDanD_4/1/console
Jenkins / Atharva_AsDanD_4 / #1
Status
Changes
Console Output
Pipeline Overview
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Copy
View as plain text
Started by user Atharva Gokhale
[Pipeline] start of pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/Atharva_AsDanD_4
[Pipeline] stage
[Pipeline] {(hello)
[Pipeline] echo
Hello, Atharva Gokhale!
[Pipeline] // stage
[Pipeline] //
[Pipeline] // node
[Pipeline] end of pipeline
Finished: SUCCESS




CONCLUSION: LO1, LO3

In this assignment, we learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.